A Preliminary Study on an Intrusion Detection Method using Large Language Models in Industrial Control Systems

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***Abstract*—In recent studies, there has been a growing interest in leveraging large language models (LLMs) to detect cyber threats and intrusions. These models, designed with an intrinsic understanding of context, exhibit the capability to identify and address emerging threats, even in cases where they haven't been explicitly trained on them. In this paper, we embark on fundamental research aimed at detecting cyber intrusions within Industrial Control System (ICS) environments using LLMs. Initially, we gather diverse data from MITRE *ATT&CK ICS framework* (Adversarial Tactics, Techniques, and Common Knowledge for ICS framework), including tactics, techniques, and open datasets. Subsequently, we organize this data into a structured learning format based on the *ATT&CK ICS matrix*. This curated learning data serves as input for training an intrusion detection model utilizing LLMs.**

***Index Terms*—Operation Technology, Industrial Control System, Large Language Model, Tactic, Technique**

I. Introduction

The landscape of operational technology (OT) and industrial control systems (ICS) environments has undergone a significant evolution from the closed networks of the past. These systems are now interconnected with external networks, facilitating access to crucial components such as Programmable Logic Controllers (PLCs), Field I/O systems, Remote Terminal Units (RTUs), among others. This integration with external networks offers a plethora of conveniences, including cloud-based remote monitoring, remote access control, energy management, and various other services [1].

However, devices such as PLCs, Field I/O systems, RTUs, commonly used in most OT/ICS environments, were developed for communication and automation purposes without considering security aspects [2]. As a result, the connection between OT/ICS environments and external networks increases the attack surface of these devices, leading to a rise in cyber-attacks targeting various industrial facilities in sectors such as water supply, power, oil, transportation, and others, starting with the Stuxnet incident in 2010 [3]. Cyber-attacks targeting OT/ICS environments can have physical implications, as shown in Table I, and unlike IT environments, they can result in significant economic and human casualties [3]–[6].

Therefore, defending against cyber-attacks and protecting organizational systems is crucial. MITRE provides methods for effectively detecting and mitigating various attack methods by analyzing attackers' tactics and techniques based on the Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) framework [7]. Organizations can effectively detect and respond to cyber-attacks through the methods provided by MITRE.

TABLE I
Well-known Cybersecurity Attacks on ICS Infrastructures

Year	Target(Victim)	Attack vector	Damage scale
2010	Iran's Nuclear Centrifuges (PLCs)	'Stuxnet' malware infection	Ruining almost one-fifth nuclear centrifuges
2016	Energy facilities in Ukranie	'BlackEnergy3' malware infection	Blackouts across western Ukraine
2017	Saudi Petrochemical plant	'Triton' malware infection	Disabling safety instrumented Systems (SIS)
2020	Honda manufacturing plants	'EKANS' ransomware infection	Multiple plants shutdowns

In recent years, studies have begun to explore using Large Language Models (LLMs) for cybersecurity. Ferrag et al. [8] presented an architecture that utilized a dataset called Edge-IIoTset and the BERT model for cyber threat detection in IoT networks. Ali et al. [9] proposed HuntGPT combined with GPT-3.5 Turbo agent to detect threats in IT infrastructure. Garaza et al. [10] used LLMs to evaluate whether MITER ATT&CK framework included knowledge information related to both threats and strategies to mitigate the threats. This paper conducts a preliminary study on LLM-based intrusion detection methods using tactics and techniques of '*MITRE ATT&CK for ICS matrix*' (hereby referred to as *ATT&CK ICS matrix* in this paper) and open datasets, excluding the

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traditional signature information (such as IP, Port, Protocol) commonly used in existing intrusion detection methods.

II. Background

A. MITRE ATT&CK for ICS

The MITRE ATT&CK framework is a standardized knowledge base reflecting real cyber-attack cases, categorizing the malicious behaviors employed by attackers into different attack groups. It organizes these groups into tactics and techniques, providing a different perspective from the traditional Cyber Kill Chain commonly used in cybersecurity. By systematizing tactics and techniques, it aims to enhance the capability to detect and respond to sophisticated attacks. ATT&CK ICS specifically classifies the behaviors of attackers as they progress towards their ultimate objectives through the IT environment. The following is a brief explanation of tactic, technique, and matrix to understand ATT&CK ICS.

- Tactic: It represents the "why" of an ATT&CK technique, indicating the objective behind the attack. It also serves as a categorizing role for each technique based on the situation. In the ICS environment, there are 12 tactics such as Initial Access, Execution, Persistence, Privilege Escalation, depending on the objectives of the attack.
- Technique: It refers to how attackers achieve tactical objectives through their actions. It allows understanding what attackers are trying to obtain and the consequences of the attack. In the industrial control system environment, there are 81 techniques such as Automated Collection, Drive-by Compromise, Denial of View, etc.
- Matrix: It visualizes the concept and relationship between tactic and technique. Each tactic encompasses various techniques, and each tactic can be utilized diversely depending on the attack objectives.

B. Cybersecurity based on Large Language Models

LLMs are models used to generate, summarize, interpret, and predict new content through data collection. LLMs are utilized in various fields such as image, text, and audio generation, and recently, research has been conducted on using them for cyber-attack detection and response [11].

LLMs are effective in analyzing unstructured data from sources such as vulnerability analysis and network traffic, utilizing deep learning and Natural Language Processing (NLP) techniques [12]. As of February 2023, GPT-3 reported 213 security vulnerabilities found in code repositories(with 4 being false positives), whereas a commercial tool detected only 99 vulnerabilities [11]. In 2023, at IEEE S&P, Hammond Pearce et al. [13] conducted a comprehensive investigation using various LLMs, evaluating synthesized scenarios, self-made scenarios, and real-world security scenarios. The results were highly positive, with LLMs successfully addressing all scenarios.

III. LLM-based Intrusion Detection Model for ICS

In this section, we propose a method to a LLM for detecting intrusions in OT/ICS environments based on MITRE ATT&CK ICS framework, as shown in Figure 1.

TABLE II
Examples of collected techniques from real cyber attacks

Real Attack E.g.	Used techniques
Stuxnet	Command-Line Interface, Hooking, I/O image, and 22 ohters
PLC-Blaster	Change Operating Mode, Denial of Service, Native API, and 5 ohters

A. Data Collection

At the data collection stage, we gather information on real-world examples of cyber attacks in OT/ICS environments using ATT&CK and other open sources such as GitHub, Google, and CISA. We focus on collecting tactics, techniques, and procedures (TTPs) used in real-life examples of cyber attacks. Table II provides those examples.

The number of Techniques used in actual attacks varies. This difference may occur due to ongoing analysis of the attacks, cases where not all analysis information has been disclosed, and instances where updates have not been made yet. In the data collection stage, it is important to investigate attacks and collect data through various open sources.

B. Data Pre-processing

At the data pre-processing stage, real attack examples are organized based on the *ATT&CK ICS matrix*, as shown in Figure 2. The organized information allows us to identify the tactics and techniques used by each attack and understand the common tactics and techniques utilized in the attacks on OT/ICS environments. For the Stuxnet malware attack, Figure 2 maps the collected data to tactics and techniques from the viewpoint of the *ATT&CK ICS matrix*. The mapped information is used to train a LLM for detecting intrusions.

C. Modeling

At the modeling stage, we build a LLM using the pre-processed data for detecting attacks. LLMs such as ChatGPT, Bard, and LLaMA can be pre-trained used through transfer learning. Fine-tuning is conducted for efficiently and correctly detecting cyber attacks. During fine-tuning, the model is trained based on the data collected in the data pre-processing stage. By analyzing and learning patterns and structures of attackers from datasets containing tactics and techniques, the model can predict potential attacks.

Within near future, we will build a LLM which can receive queries and generate answers. Using the LLM, a defender can prepare for upcoming attacks and her or his organizations can understand the intentions of attackers and minimize the possibility of security incidents. An example of a query and its response between the user and the model is as follows.

- User: Please tell me about techniques that may occur in PLC.
- Model: Techniques that can occur in PLC are as follows. Replication Through Removable Media, Supply Chain Compromise, Rogue Master, etc.
- User: So far, Supply Chain Compromise and Rogue Master techniques have occurred in the Initial Access stage in PLC devices. Let me know about the next possible attacks.

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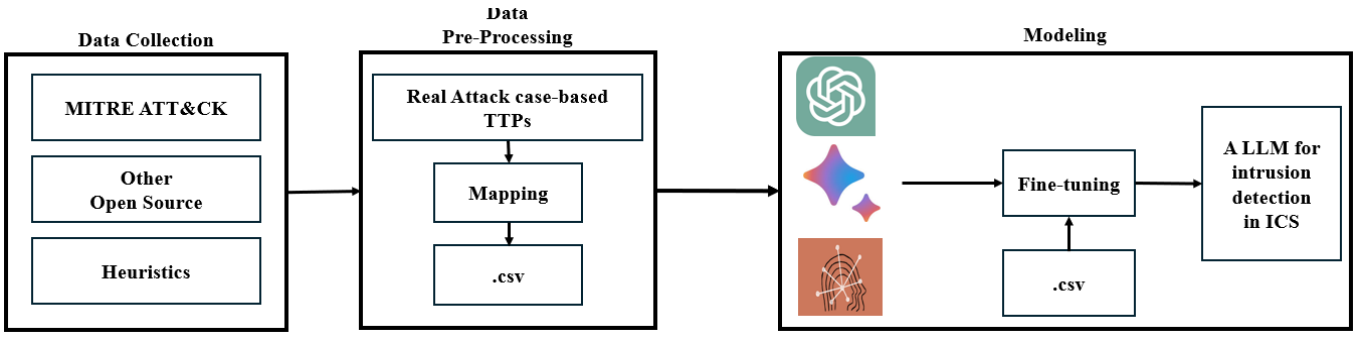


Fig. 1. A Structure to Build a LLM for Intrusion Detection in OT/ICS



Fig. 2. Tactics and Techniques related to Stuxnet Malware Attack

- Model: The next expected potential tactic is Execution. Attacks such as Change Operating Mode (55%), Execution through API (30%), and Modify Controller Tasking (10%) may occur.

IV. Conclusion and Discussion

In the realms of Operational Technology (OT) and Industrial Control Systems (ICS), the application of intrusion detection techniques from Information Technology (IT) proves challenging due to the distinctive characteristics of OT and ICS devices and protocols. Moreover, conventional intrusion detection methods heavily reliant on signature information such as IP addresses, source-destination pairs, MAC addresses, etc., are susceptible to alteration by attackers [14], thus complicating detection and response processes should attackers successfully evade detection methods [15].

In response to these challenges, we have introduced an approach to intrusion detection tailored specifically for OT/ICS environments utilizing LLMs. This proposed technique represents foundational research aimed at conceptual establishment. The envisaged methodology will undergo prototype development and subsequent testing in the future.

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